

FACTORS ASSOCIATED WITH DISTRESS FOLLOWING THE BREAKUP OF A CLOSE RELATIONSHIP

Susan Sprecher

Illinois State University

Diane Felmlee

University of California — Davis

Sandra Metts

Illinois State University

Beverley Fehr

University of Winnipeg

&

Debra Vanni

Illinois State University and Information Resources, Inc.

ABSTRACT

The purpose of this investigation was to identify the factors associated with the distress experienced after the breakup of a romantic relationship, both at the time of the breakup (assessed retrospectively) and at the time the questionnaire was completed. Four categories of variables were examined as possible correlates of post-breakup distress: variables associated with the initiation of the relationship, characteristics of the relationship while it was intact, conditions at the time of the breakup and individual difference variables. The sample consisted of 257 young adults (primarily college students; 83 male and 174 female) who had experienced a recent breakup ($M = 21$ weeks since breakup). The variables most highly associated with distress at the time of the breakup were non-mutuality in alternatives (i.e. partner having more interest in alternatives), commitment, satisfaction, greater effort in

For correspondence on this paper, contact the first author at: Department of Sociology and Anthropology, Illinois State University, Normal, IL 61790-4660, USA.
[email: sprecher@ilstu.edu.]

Journal of Social and Personal Relationships Copyright © 1998 SAGE Publications (London, Thousand Oaks, CA and New Delhi), Vol. 15(6): 791-809. [0265-4075 (199812) 15:6; 005565]

relationship initiation, being 'left' by the other and fearful attachment style. The variables most highly associated with current distress were commitment, duration of the relationship, fearful attachment style, dismissing attachment style and time since breakup.

KEY WORDS • break up of close relationship • distress

The breakup of a close relationship can be one of life's most emotional experiences, although there seems to be considerable variability in the level of overall distress, the specific emotions experienced (e.g. guilt, anger, sorrow), and the duration of the emotional distress that is experienced (Frazier & Cook, 1993; Sprecher, 1994). As several emotion theorists (e.g. Lazarus et al., 1970) have argued, emotions are adaptive responses to social events. When the stimulus prompting the emotional response is the dissolution of affectional ties, the experience of emotional distress depends upon how an individual construes the various features of the situation that are salient at the time of the breakup and subsequently.

Although relationship researchers have begun to identify factors that might influence the nature of post-dissolution distress, most investigations have considered relatively few factors at a time. The three major categories of variables that have been examined are (i) characteristics of the relationship while it was intact, (ii) conditions under which the relationship ended, and (iii) individual difference variables. Because these categories of variables have typically not been examined in conjunction, the relative contribution to post-dissolution distress of these categories of variables is unknown.

The present investigation examines, within a multivariate framework, the degree to which a variety of factors representing different stages in the life-course of a relationship are associated with the distress experienced after a breakup. We consider variables within each of the above categories as well as a fourth set of variables not typically studied in conjunction with relationship dissolution: factors associated with the initiation of the relationship. Young adults who had experienced the dissolution of a dating relationship during the previous year completed a questionnaire about the breakup experience, including their emotional reactions and subjective ratings of distress, both at the time of the breakup and currently.

Although *factors associated with the initiation of the relationship* have not generally been linked to factors associated with its ending, there are at least two exceptions. Metts et al. (1989) found that several factors, including being friends before the development of a romantic relationship, were associated with the likelihood that ex-partners remained friends after the breakup. Felmlee (1995) found that the qualities that are initially attractive in a romantic partner (e.g. 'successful') are sometimes closely related to characteristics that are eventually disliked (e.g. 'workaholic'), and can contribute to relationship dissolution. Thus, it is reasonable to expect that

factors associated with the initiation of the relationship might also be associated with the distress experienced after the breakup.

One variable associated with the genesis of a relationship that is likely to be related to post-breakup distress is the effort put into developing the relationship. We hypothesize that a person who exerts a great deal of effort to initiate a relationship that eventually terminates will feel more distress than the individual who has exerted less effort. This effect may occur even controlling for the initiation of the breakup (presumably the person who initiates the relationship may be less likely to initiate the breakup) and can be explained from either a dissonance (e.g. Brehm & Cohen, 1962) or investment (Rusbult, 1983) perspective. That is, a person who invests a great deal into launching a relationship may feel a high level of distress (and dissonance) because of the effort he or she invested in an unsuccessful relationship.

The number of reasons promoting initial attraction also may be associated with post-dissolution distress. Some relationships are 'matches made in heaven'. The partners are attracted to each other for a number of reasons, for example, physical attractiveness, similarity across a number of domains, proximity, sharing friends in common, and so forth. In other matches, the attraction may be based on only one or a few factors. We expect that it will be more distressing to experience the dissolution of a liaison one forms with another because of multiple sources of attraction.

The *characteristics of a relationship while it was intact* also may forecast the distress experienced after a breakup. In particular, satisfaction and commitment in the relationship before the breakup and the duration of the relationship are variables likely to be related to post-dissolution distress. These variables are components of interdependence theory (e.g. Kelley & Thibaut, 1978) and also are included in an extension of this theory, Rusbult's (1983) investment model. Although the interdependence/investment framework has been used most often to forecast the stability of relationships, it also has been used to predict the degree of distress experienced by those who break up. According to this framework, people who are more invested (e.g. have been together longer), committed, satisfied in their relationship, and who have fewer alternatives should be more distressed if the relationship breaks up. These predictions have received empirical support. For example, several researchers have found that the longer the relationship, the greater the distress experienced when it ends (e.g. Attridge et al., 1995; Berscheid et al., 1989; Fraley et al., 1997). Furthermore, a positive association has been found between satisfaction (and love) in a relationship and post-breakup distress (e.g. Attridge et al., 1995; Fraley et al., 1997).

Support for interdependence theory predictions also has been found when these variables are studied in combination. For example, in a 3-month longitudinal study, Simpson (1987) found that among dating individuals who experienced a breakup, closeness, duration of relationship, and ease of finding a partner (all measured before the breakup) were associated independently with the distress experienced after the breakup.

Similarly, Attridge et al. (1995) found that variables such as commitment, love, investment, dependence and poor alternatives predicted post-breakup distress, although only for men. Frazier & Cook (1993) found that satisfaction, closeness and perceived alternatives were related to retrospective ratings of stressfulness immediately after the breakup. However, these variables were generally unassociated with current feelings of recovery. Finally, in a recent longitudinal study, Fine & Sacher (1997) found that commitment (measured while the relationship was intact) predicted level of distress after the breakup for both men and women, and that satisfaction predicted level of distress only for men.

The *conditions under which a relationship ends* are proximate to the breakup and thus may be most highly associated with the post-breakup distress. The factor that is most likely to be related to the distress level is who initiated the breakup. Most relationship breakups are non-mutual (e.g. Hill et al., 1976; Sprecher, 1994); one partner wants to end the relationship more than the other. Research indicates that the person who initiates the breakup is less distressed than the person who is left, at least at the time of the breakup (Attridge et al., 1995; Frazier & Cook, 1993; Sprecher, 1994). The different emotional experiences of the 'leaver' vs the 'left' are attributable, in part, to the fact that life events perceived to be under one's control are less distressing than those perceived to be beyond one's control (Fiske & Taylor, 1984).

However, the same psychological benefits may not be associated with another domain of control potentially experienced at the time of the breakup, which is perceiving oneself as responsible for the problems leading to the breakup. Whereas the initiation of a breakup may be desirable control, perceiving oneself as responsible for the problems leading up to the breakup may be undesirable control (Gray & Silver, 1990), and is likely to be associated with distress. In partial support of this, Sprecher (1994) found a trend ($p < .10$) for perceived responsibility for the breakup to be associated with post-breakup distress among a sample of ex-daters.

One other condition existing at the time of the breakup that is likely to be associated with post-dissolution distress is the degree of non-mutuality of alternatives available to the partners. Some people have a desirable alternative (or alternatives) available to them at the time of the breakup, whereas others do not. Although this condition may have existed as a characteristic of the relationship while intact (and is often studied as such), it becomes especially salient when (i) it is a factor that contributes to the breakup of the relationship, either perceptually or in fact, or (ii) the availability of an alternative lessens the sense of loss for one partner but exacerbates the sense of loss for the partner who does not perceive alternatives as available. Research supports the association between both of these conditions and distress after a breakup. For example, research indicates that when the dissolution of a relationship is attributed to partner's interest in someone else, a high level of distress is likely, whereas one's own interest in someone else is associated with a low level of distress

(Sprecher, 1994). Similarly, Drigotas (1996) found that unavailability of attractive alternatives was associated with post-breakup distress and Fraley et al. (1997) found that involvement in a new relationship was positively correlated with emotional recovery.

Characteristics of the individuals may also be associated with the degree of distress experienced after the breakup. That is, we should expect that, even under similar conditions, some individuals will experience greater upset over a breakup and endure a longer period of recovery than will other individuals, perhaps because of differential coping resources and/or differential tendencies to ruminate or obsess over failures (e.g. McIntosh & Martin, 1992). Furthermore, the distress felt at the time of the breakup and experienced during the months that follow, might be greater for individuals who are generally insecure about their ability to establish and maintain enduring relationships.

More specifically, *adult attachment style* has been found to be associated with a number of relationship phenomena (for a review, see Hazan & Shaver, 1994), including relationship dissolution (e.g. Simpson, 1990). Because of different relationship histories and dispositions, individuals with different adult attachment styles are not likely to have similar reactions to the breakup of a close relationship, although the particular associations found between attachment styles and post-dissolution distress vary across research studies and attachment style measures. Feeney & Noller (1992) found a positive relation between ambivalent attachment style and upset after a breakup, but a negative relation between avoidant attachment style and upset (the secure attachment style was unrelated to post-dissolution distress). Simpson (1990) also found that the avoidant attachment style was associated with less emotional distress after a breakup, but only for men; in the Simpson (1990) study, secure and anxious styles were unrelated to post-dissolution distress. In a more recent study, Fraley et al. (1997) found that individuals who were high on Anxiety (a dimension that combined preoccupied and fearful attachment styles) experienced higher levels of denial, protest and despair when a relationship ended (even when controlling for relationship satisfaction) than those low in Anxiety.

A second individual difference variable frequently studied in previous research on breakup distress is *sex of respondent*. Hill et al. (1976) found that men were more distressed than women after the breakup of a dating relationship. However, more recent investigations have found no sex differences in overall distress experienced after the breakup of a dating relationship (e.g. Frazier & Cook, 1993; Simpson, 1990; Sprecher, 1994).

The present investigation extends previous research on factors associated with post-breakup dissolution in several ways. First, we take a relationship stage approach, including an examination of conditions under which the relationship began as well as conditions before its dissolution. Second, we examine diverse correlates of post-dissolution distress within a multiple regression framework. Because many predictors of breakup distress are likely to be intercorrelated, the unique contribution of various

predictors to distress needs to be determined. Some variables found to be important in previous research may not be significant predictors once other variables, perhaps more proximal to the breakup, are controlled. Third, the present study also extends previous research by examining both initial distress following the breakup and current distress. Several recent studies have shown that the amount of time elapsed since the breakup is correlated with less distress (e.g. Attridge et al., 1995), greater emotional recovery (Fraley et al., 1997), and more positive interaction with the former partner (Drigotas, 1996). Thus, reactions to a breakup immediately after the event may differ from those at a later time. Perhaps more important, the variables that predict degree of initial distress may differ from the variables that predict lingering distress in the months after the breakup has occurred (see, for example, Frazier & Cook, 1993). In order to determine possible differences in patterns of variables associated with distress immediately following a breakup compared with distress in the months following the breakup, we asked participants to report on both initial distress (retrospective assessments) and current distress (at the time of data collection). We also included two measures of initial distress — a global item of upset and a detailed measure of a variety of negative and positive emotions.

Our specific hypotheses are that post-breakup distress will be associated with the following variables (ranging from most distal to most proximal). For relationship initiation variables, we predict that self-initiation of the relationship and multiplicity of reasons for early attraction will be associated with greater distress. With regard to characteristics of the relationship while it was intact, we expect that higher satisfaction and commitment, as well as greater relationship duration, will be associated with greater distress. For variables related to the breakup circumstances, we expect that partner's initiation of the breakup, perceived self-responsibility for problems leading to the breakup, and perceiving that the partner had more interest in an alternative will be associated with greater distress. Finally, we expect that attachment style, specifically insecure types of attachment (i.e. preoccupied and fearful) will be associated with increased distress, but that men and women will not differ in level of distress.

Regarding differential effects on immediate and subsequent breakup distress, we predict that people will be more upset about the breakup immediately after the breakup than currently, and that the variables we have identified as factors associated with post-breakup distress will be more highly associated with initial distress than with current distress. Of the four categories of variables hypothesized to be associated with post-breakup distress, we expect that conditions proximal to the breakup (e.g. control over the initiation) will be most highly predictive of post-breakup distress, particularly immediately after the breakup.

Method

The *sample* consisted of 257 young adults (83 men and 174 women) who had experienced the breakup of a close relationship within the previous 12 months.

An additional 33 participants were in the original sample but were not included in the analyses because they completed the questionnaire about a relationship that had ended more than 12 months before the completion of the questionnaire ($n = 12$) or because they represented an outlier in another way (we eliminated eight participants who were over the age of 30, four additional participants who completed the questionnaire about a divorce, and nine who completed the questionnaire for a non-romantic/non-dating relationship). The mean age of the final sample was approximately 20 years; most were students at a public midwestern university. Although information on racial/ethnic background of the sample was not obtained, the undergraduate population of the university was 86.4 percent White, 8.2 percent Black and 5.4 percent other racial/ethnic backgrounds.

The participants were obtained through a network sampling procedure. Students in two classes taught by the first author in spring 1995 were presented, as an optional research assignment, the task of distributing a questionnaire and conducting a follow-up interview with a person from their social network who had experienced the breakup of a close relationship during the past 12 months. (The follow-up interview was used for instructional purposes only; these data were not collected.) Students were given oral and written instructions on how to administer the questionnaire to their selected participant. In addition, the participants received a letter from the investigator explaining the purpose of the study and instructions to return the questionnaire in a sealed envelope either directly to the student assistant or through campus mail to the instructor. The voluntary nature of the study and the anonymity/confidentiality of the data were emphasized to the participants.

The *questionnaire* directed the participants to think of a person with whom they 'had a close and very involved relationship for a while, but the relationship has since ended. This could be a romantic partner or a very special friend. The person could be of either the same sex or the opposite sex'. (Because most participants had selected a romantic partner, as explained earlier, we eliminated from the analyses the few who selected the breakup of a friendship.) The questionnaire had two major sections: (i) measures of factors that led to their early attraction for their ex-partner, and (ii) measures of various aspects of the breakup. We focus in this paper on the latter set of measures, with the exception of a composite score created from the attraction measures, which indicates the number of predictors of attraction that were rated as important.

Factors associated with the initiation of the relationship. Effort put into the initiation of the relationship was assessed by the question, 'Who would you say worked harder to initiate the relationship?'. Seven possible responses followed: 1 = The other did a lot more, 2 = The other did somewhat more, 3 = The other did a little more, 4 = We both did equally, 5 = I did a little more, 6 = I did somewhat more, 7 = I did a lot more. Multiplicity of reasons for attraction was represented by a count of the number of predictors of attraction (out of 17: e.g. other's physical attractiveness, similarity on background characteristics, similarity on attitudes and values, support from significant others) that received either a 3- or 4-importance rating on a 4-point scale. Thus, the higher the score, the greater the multiplicity of reasons.

Characteristics of the relationship while intact. Satisfaction in the relationship (before the breakup) was measured retrospectively by the question, 'Before the relationship ended, how satisfying was it?' (1 = not at all, to 7 = extremely). Similarly, commitment was measured by the question, 'Before the

relationship ended, how committed were you to maintaining the relationship in the future?' (1 = not at all, to 7 = extremely). Duration of the relationship was assessed by the question, 'How long had you been in a close relationship with this person when the relationship ended?' (participants were instructed to respond in years and/or months and responses were recoded in months).

Conditions under which the relationship ended. Control over the initiation of the breakup was assessed by the item: 'Think about who actually made the first steps to initiate the breakup. Circle the number that best represents who initiated the breakup'. A 7-point response scale followed, with the anchors 1 = I did; 4 = We both did; and 7 = My partner did. Control over responsibility for the breakup was assessed by the item, 'Think back to the problems that led up to the breakup. Circle the number that best represents who you think contributed more to the problems' (a similar 7-point response scale was used). Both items were recoded so that the higher number indicated more self-control. Non-mutuality of alternatives at the time of the breakup was assessed using two items from a larger list of reasons (see Sprecher, 1994) for ending of the relationship (each reason was rated on a 1 = not at all important; to 7 = extremely important response scale). Participants' response to the item, 'I became interested in someone else' was subtracted from their response to the item, 'My partner became interested in someone else'. The higher the number, the greater the non-mutuality of alternatives in the direction of the partner being perceived as having more interest in an alternative at the time of the breakup.

Attachment variables. The paragraph descriptions of the four attachment types (secure, fearful-avoidant, preoccupied and dismissing-avoidant) from Bartholomew & Horowitz (1991) were provided to the participants. Each paragraph was followed by a 7-point response scale ranging from 1 = not at all true of me, to 7 = very true of me.

Measures of distress experienced after the breakup. A detailed measure of the distress experienced immediately after the breakup was derived from participants' responses to items that assessed the degree to which they recalled experiencing certain emotions right after the breakup. The particular emotions were chosen from a list used in another study of relationship breakups (Sprecher, 1994). The nine negative emotions were depression, guilt, anger, hate, frustration, resentment, loneliness, jealousy and hurt. The five positive emotions were contentment, love, happiness, satisfaction and relief. A 7-point response scale (1 = not at all; 4 = somewhat; 7 = extremely) was provided for each emotion. From the responses to these emotions, an Overall Distress Index was created based on the difference between the mean response to the negative emotions and the mean response to the positive emotions. A higher, positive score indicated that negative emotions were experienced to a greater degree than positive emotions. A similar index, sometimes called a Hedonic Emotional Tone Index (Berscheid et al., 1989), has been used in previous research on emotions. Although some may argue that a post-dissolution distress measure should contain only negative emotions, we assume that a more valid measure focuses on both the presence of negative emotions and the absence of positive emotions. To calculate a reliability index, we reverse scored the positive emotion items and conducted coefficient alpha on the combined negative and positive items. This reliability coefficient was .84.

After responding to the questions on their specific emotional reactions after the breakup, participants were asked 'Overall, how upset were you after the

breakup?' A 7-point response scale was provided (1 = not at all, 4 = somewhat, 7 = extremely). This measure represents a Global Measure of Initial Upset. This question was followed by a Global Measure of Current Upset: 'Overall, how upset are you now about the breakup?' (the same response scale was provided).

The three measures of distress were positively interrelated. The Overall Distress Index was correlated .76 with the Global Measure of Initial Upset, and .46 with the Global Measure of Current Upset. The correlation between the two global measures was .51 (all significant at $p < .001$).

Results

Of the participants included in the final analysis, most (98%) described an opposite-sex relationship. Descriptive statistics for most of the variables are presented in Table 1.

Both negative *and* positive emotions were experienced immediately after the breakup, although the mean representing the degree to which the nine negative emotions were experienced was greater than the mean representing the degree to which the five positive emotions were experienced ($M = 4.20$ vs 3.33 ; $t(247) = -5.91$, $p < .001$). Hence, the overall distress index was positive. (A table reporting the mean response to the specific emotions is available to those who contact the first author.) As expected, participants scored higher on the initial upset item than on the current upset item ($t(252) = 18.57$, $p < .001$).

The mean on the initiation of relationship item was near the mid-point for men, indicating equal initiation. However, women had a significantly lower score (in the direction of partner initiation) than did men. The participants reported becoming attracted to their partner for a number of reasons; for both men and women, almost 11 factors (out of 17) were rated as somewhat important, or the major reason for attraction.

The participants reported they had been very satisfied in the relationship and committed to maintaining it prior to the breakup of the relationship. The relationships had ranged in duration from 1 month to 156 months, with a mean of 22 months and a median of 18 months.

According to the participants' reports, only 12 percent indicated that the breakup was entirely mutual (i.e. chose the response indicating equal initiation). Fifty-two percent said they took more initiative to end the relationship, and 36 percent said that their partner took more initiative (the mean response to this item was in the direction of self-initiation). To the question asking about who was responsible for the problems that led up to the breakup, 30 percent indicated that both contributed equally to the problems, 50 percent said their partner contributed more, and 20 percent said that they contributed more (the mean response to this item was in the direction of partner responsibility). Participants perceived that the partner had slightly more interest than the self in alternatives. The duration between the time of the breakup and the current period ranged from 1 week to 52 weeks (mean = 21 weeks).

Each attachment style described the participants to some degree. However, most characteristic of the participants was the secure attachment style and least characteristic of them was the preoccupied style.

The major analyses centered on the question of how variables from each category of the life-stage of the relationship are related to the distress experienced after a breakup. Table 2 presents correlations among the predictor

variables included in these analyses for the total sample. Each variable was associated significantly with two or more of the other variables, although none of the correlations was more than moderate in size. Table 3 presents the zero-order correlations between the categories of variables and the three measures of distress after the breakup. All the variables were associated with one or more of the distress measures in the bivariate analyses. Below, we discuss only the effects found to be significant in the multivariate analyses, which are presented in Table 4.

Relationship initiation variables. The perception of how much effort self vs other had put into the initiation of the relationship was significantly associated with distress after the breakup. The greater one's own effort into the initiation of the relationship, the greater was one's initial distress after the breakup (measured by both the Distress Index and the Initial Upset item). However,

TABLE 1
Descriptive statistics on the variables

	Mean for total sample (<i>N</i> = 257)	Mean for men (<i>n</i> = 83)	Mean for women (<i>n</i> = 174)
Distress after breakup			
Overall distress index ^a	.87	.66	.97
Initial upset ^b	5.17	4.70	5.40**
Current upset ^b	3.07	2.78	3.21
Variables associated with the initiation of the relationship			
Initiation of the relationship ^c	3.46	4.02	3.19***
Multiplicity of reasons for attraction ^d	10.73	10.77	10.71
Relationship characteristics (interdependence variables)			
Satisfaction ^b	4.96	5.31	4.79*
Commitment ^b	5.25	5.00	5.37
Duration	22.36 (mos)	22.36 (mos)	22.36 (mos)
Conditions at the time of the breakup			
Initiation of breakup ^c	4.40	3.77	4.70**
Responsibility for breakup ^e	3.26	3.62	3.09*
Non-mutuality of alternatives ^f	.27	.39	.21
Time since breakup	20.90 (wks)	21.53 (wks)	20.60 (wks)
Individual difference variables (attachment types)			
Secure attachment style ^b	4.23	4.23	4.24
Fearful attachment style ^b	3.94	3.55	4.12*
Preoccupied attachment style ^b	3.48	3.00	3.70**
Dismissing attachment style ^b	3.84	3.84	3.83

Notes: The *N* size for the analyses conducted on the total sample ranged from 248 to 257, depending on amount of missing data. The *n* sizes ranged from 80 to 83 for men and 168 to 174 for women.

^a Higher numbers indicate that negative emotions were experienced to a greater degree than positive emotions.

^b Higher numbers indicate more of these variables.

^c Higher numbers indicate greater own effort relative to partner effort.

^d Refers to the mean number of items (out of 17) rated either 3 or 4 on a 4-point importance scale.

^e Higher numbers indicate more control attributed to the self relative to the partner.

^f Higher numbers indicate greater non-mutuality of alternatives in the direction of perceiving that the partner had more interest than oneself in an alternative.

Asterisks indicate that a significant difference was found between men and women, at: * $p < .05$;

** $p < .01$; *** $p < .001$.

perceived initiation of the relationship did not significantly affect current distress. Multiplicity of reasons for early attraction was also associated with distress after the breakup, as measured by the Initial Upset item.

Characteristics of the relationship (interdependence variables). Of the interdependence variables (commitment, satisfaction and duration of the relationship), commitment had the strongest association with distress after the breakup. Greater commitment was associated with greater distress after the breakup, as assessed by all three distress measures. Satisfaction was associated with greater initial upset (significant for both the Distress Index and the Global Measure of Initial Upset). Duration of the relationship had a strong and positive effect on current upset, and also a significant effect on initial upset, as measured by the global item.

Conditions at the time of the breakup. Non-mutuality of alternatives at the time of the breakup predicted degree of initial upset after the breakup, measured via both the Distress Index and the Global Measure of Initial Upset. More specifically, the greater the perceived non-mutuality of alternatives, in the direction of the partner having greater interest in an alternative, the greater the distress that was experienced. The perceived initiation of the breakup had a weaker effect on distress level, but the coefficient was significant (and negative) both for the Distress Index and the Global measure of initial upset. However, neither perceived initiation of the breakup nor non-mutuality of alternatives was associated with distress experienced currently. Furthermore, perceived responsibility for the problems leading up to the breakup was not associated with any of the three measures of distress after the breakup in the multivariate analyses. We had also included length of time since breakup, primarily as a control variable. Time since breakup had no effect on the level of distress reported initially after the breakup (on either the Distress Index or the Initial Upset item), which suggests that how people recall their distress for the period of time after the breakup is unaffected by the passage of time (i.e. suggests a memory bias was not occurring). However, as would be expected, time since breakup did affect the level of distress reported currently. The longer the period since the breakup, the less the current distress.

Individual difference variables. In the multivariate analysis, one attachment type, fearful, was consistently and positively associated with distress level. Higher scores on the fearful attachment style were associated with more distress experienced after the breakup, and the effect of this measure was significant for all three distress measures. Furthermore, the dismissing attachment style was significantly associated with a lower level of current upset. We also conducted the analyses with the four attachment responses reduced to two theoretically independent dimensions (Anxiety = (Preoccupied + Fearful) – (Secure + Dismissing); Avoidant = (Fearful + Dismissing) – (Secure + Preoccupied)), as recommended by some attachment researchers (e.g. Fraley et al., 1997). Both dimensions were found to be positively associated with the experience of distress after the breakup in zero-order correlations (Anxiety: $r = .36$ with Distress index; $r = .28$ with Initial Upset; and $r = .27$ with Current Upset; all $p < .001$; Avoidance: $r = .22$ ($p = .001$) with Distress index; $r = .16$ ($p = .01$) with Initial Upset; and $r = .15$ ($p = .02$) with Current Upset). When scores on these two dimensions were included in simultaneous regression with all other predictors, the Anxiety dimension was a significant, positive predictor of each of the distress measures, whereas the Avoidance dimension was not. The results for the other predictors were generally unaffected.

TABLE 2
Zero-order correlations among the predictor variables

	INR	MUR	SAT	COM	DUR	INB	REB	NMU	TIM	GEN	SEC	FEA	PRE	DIS
Initiation of relationship (INR)														
Multiplicity of reasons for attraction (MUR)	.00													
Satisfaction (SAT)	.01	.11*												
Commitment (COM)	.10*	.21***	.44***											
Duration (DUR)		.22***	.12*	.16**										
Initiation of breakup (INB)														
Responsibility for breakup (REB)														
Non-mutuality of alternatives (NMU)														
Time since breakup (TIM)														
Sex (GEN)														
Secure attachment style (SEC)														
Fearful attachment style (FEA)														
Preoccupied attachment style (PRE)														
Dismissing attachment style (DIS)														

Note: The exact N size for the analyses ranged from 251 to 256, depending on amount of missing data.
* $p < .05$; ** $p < .01$; *** $p < .001$.

Respondent sex was associated significantly with only one distress measure, the global item of initial upset. Women reported more initial distress than did men. (The means for men vs women on this item are reported in Table 1.)

Discussion

This study explored the unique contribution of a number of factors to the experience of distress immediately following and subsequent to the breakup of a dating relationship. The findings are generally consistent with previous research and our predictions, and they provide support for our multivariate conception of breakup distress.

We find that the amount of distress an individual experiences immediately following a breakup is associated significantly with predictors from each of our four categories of independent variables. For variables associated with the *initiation of the relationship*, findings reveal that a sense of distress is greater at the time of a breakup if more effort was made to initiate the relationship. Concerning *characteristics of the relationship while it was intact*, results indicate that post-dissolution distress is also relatively high when the relationship is characterized by high commitment, satisfaction and duration. In addition, for variables associated with *conditions at*

TABLE 3
Zero-order correlations between predictors variables and the distress measures

	Distress index	Initial upset	Current upset
Variables associated with the initiation of the relationship			
Initiation of relationship	.15**	.18**	.05
Multiplicity of reasons for attraction	.15**	.25***	.10*
Relationship characteristics (interdependence variables)			
Satisfaction	.30***	.28***	.16**
Commitment	.51***	.56***	.36***
Duration	.05	.14**	.21***
Conditions at the time of the breakup			
Initiation of breakup	-.39***	-.32***	-.10*
Responsibility for breakup	-.13*	-.15**	.01
Non-mutuality of alternatives	.47***	.38***	.15**
Time since breakup	.04	-.01	-.28***
Individual difference variables			
Secure attachment style	-.12*	-.05	-.07
Fearful attachment style	.28***	.19***	.22***
Preoccupied attachment style	.22***	.21***	.15**
Dismissing attachment style	-.18**	-.16**	-.18**
Sex of respondent	.06	.18**	.10*

Note: The exact *N* size was 243 for the correlations of the predictor variables with the Distress index, 250 for the correlations with Initial upset, and 248 for the correlations with Current upset. The *N* size varied because of varying amounts of missing data.

* $p < .05$; ** $p < .01$; *** $p < .001$.

the time of the breakup, findings suggest that a breakup may be most distressful when one is 'left' for another person. Similar conclusions have been reached in related research (e.g. Frazier & Cook, 1993; Sprecher, 1994).

Finally, the two types of *individual differences* variables, attachment style and sex of respondent, yielded only one consistent pattern: having a fearful attachment style (or scoring high on the anxiety dimension of attachment) predicts both initial distress and subsequent distress. Sex of respondent was a significant predictor of initial distress only on the global item with females reporting significantly more upset than males. Although speculative, women in this sample may have experienced greater distress at the ending of their relationships than men because they were also more likely than men to attribute the problems leading to breakup to their partners (see Table 1). Of course, it also may be that females find it easier to admit experiencing negative emotions than males. Nevertheless, differences between males and females appear to be temporary. In measures of the degree of distress experienced currently, sex differences of any kind disappear.

TABLE 4
Standardized betas for predictors of distress: results from multivariate regression

Predictor variable	Distress index	Initial upset	Current upset
Variables associated with the initiation of the relationship			
Initiation of relationship	.11*	.16**	-.02
Multiplicity of reasons for attraction	.03	.13**	.02
Relationship characteristics (interdependence variables)			
Satisfaction	.17**	.14*	.10
Commitment	.21***	.28***	.22**
Duration	.06	.10*	.20***
Conditions at the time of the breakup			
Initiation of breakup	-.13*	-.14*	-.02
Responsibility for breakup	.02	-.03	.03
Non-mutuality of alternatives	.31***	.17**	.04
Time since breakup	.06	.00	-.28***
Individual difference variables			
Secure attachment type	-.04	.00	-.04
Fearful attachment type	.24***	.14**	.18**
Preoccupied attachment type	.10	.07	.06
Dismissing attachment type	-.06	-.05	-.13*
Sex of respondent	.07	.19***	.05
Adjusted R ²	.46***	.43***	.26***

Note: The variables are grouped according to category for purposes of illustration but all variables were entered simultaneously. The exact *N* size for each regression was: 243 for Distress index, 250 for Initial upset and 248 for Current upset.

* $p < .05$; ** $p < .01$; *** $p < .001$.

As predicted, we find that people report being more distressed immediately following the breakup than they do at the time of the survey (current distress), and that the variables we targeted as predictors of breakup distress are more useful in explaining initial distress (adjusted $R^2 = .46$ and $.43$ for Distress Index and Global Upset) than subsequent distress (total $R^2 = .26$). Current upset is, nevertheless, significantly influenced by variables from all four categories of predictors, except those associated with the initiation of the relationship (i.e. the most distal variables). As a whole, the pattern of findings for current distress suggests that the common adage, 'time heals a broken heart', is only partially true. When early commitment to maintaining the relationship is strong and the relationship constitutes a large part of a person's dating history and social network (long duration), distress continues to be felt long after the breakup, especially by individuals who are dispositionally fearful of relationships.

As expected, some variables do not contribute unique variance to post-breakup distress when other variables are present in the same regression model. Responsibility for the problems leading to the breakup, for example, failed to predict distress on any measure, despite a significant zero-order correlation with distress. Apparently the more cognitive attributional process of assigning blame contributes only indirectly to the experience of distress, whereas partner's initiation of the breakup and, particularly, the belief that one's partner is interested in another person contribute independent and unique variance.

Taken together, these findings lend a good deal of support to existing theoretical accounts of relationship termination. Two theoretical frameworks are particularly germane: Interdependence Theory and Attachment Theory.

According to Interdependence Theory (e.g. Kelley & Thibaut, 1978), and its extension, the Investment Model (Rusbult, 1983), individuals who are more invested in a relationship (e.g. together longer, more committed), should experience more distress following the demise of a relationship. We found a good deal of support for this argument, and our results corroborated those of several other studies. For example, previous research has found a significant association between relationship length and breakup distress (e.g. Attridge et al., 1995; Berscheid et al., 1989; Simpson, 1987), and we found a similar association for current distress. A positive relationship was also found between satisfaction and distress experienced immediately after a breakup (for similar findings, see Attridge et al., 1995).

The most potent investment/interdependence variable in our analyses, however, is commitment. It yielded a consistently large and highly significant effect on all three measures of distress, meaning that it was related both to distress immediately following the breakup and current distress. This finding also confirms those of other studies (e.g. Attridge et al., 1995; Fine & Sacher, 1997). According to Rusbult's (1983) investment model, subjective commitment to a relationship is the most immediate predictor of relationship stability. Thus, commitment may be an especially important

indicator of distress after a relationship ends because it reflects the extent to which individuals wanted and/or expected, their relationship to continue. Highly committed individuals must therefore adapt not only to changes in their immediate environment, but also to changes in the future as they had envisioned it.

Attachment theory also receives support in our analyses. Three of the four attachment styles (Fearful, Preoccupied and Dismissing) are significantly correlated with all three measures of distress (Secure was the only exception, but it was correlated negatively with the Distress Index). In addition, having a fearful attachment style is associated with all the measures of upset, even after controlling for other variables in the multivariate analyses, and the dismissing attachment style is uniquely associated with current distress (in the opposite direction as the fearful attachment style). Similarly, Birnbaum et al. (1997) found that individuals with an anxious-ambivalent attachment style have been found to exhibit higher degrees of psychological distress associated with divorce than those with a secure style.

The variables measuring a secure, preoccupied and dismissive attachment styles failed to have significant multivariate effects on post-breakup distress, probably because these types are not independent of each other (e.g. those who score high in anxiety are likely to have a low secure and dismissing score.) When measures of two latent, theoretically independent, attachment dimensions were used in place of the four styles, the Anxiety dimension, but not the Avoidance dimension, was significantly and positively related to breakup distress in the multivariate analyses. In a related vein, Fraley et al. (1997) also found that individuals scoring high on the Anxiety attachment dimension (but not the Avoidant dimension) experienced higher levels of protest, anger and despair following a breakup, than those who were less anxious, even controlling for the investment measures of satisfaction and commitment.

One possible explanation for the higher distress levels of anxious individuals is that they use relatively poor coping techniques during a crisis (Mikulincer et al., 1993). Birnbaum et al. (1997), for example, found that in response to a divorce, anxious-ambivalent adults reported, to a greater degree than secures, more maladaptive coping mechanisms, such as the use of self-defeating thoughts, withdrawal and wishful thinking.

Importantly, our findings are not explained completely by one particular relationship theory. An interdependence perspective accounts for a good deal of variance in our models, but not all. Attachment theory explains additional variance in breakup distress, but again only a portion of the variance. In addition, other factors not associated clearly with any one major theoretical perspective are also influential in the model. One of the main strengths of this analysis, therefore, is that unlike many investigations that focus on only one approach, it demonstrates that various theoretical perspectives are needed to understand the multidimensional nature of the emotional reactions immediately following and long after a breakup.

As with any study, the contributions of the findings reported here must

be considered in light of the limitations of the research. Although a strength of this study is the relatively large sample compared with most breakup studies (e.g. both Simpson, 1987 and Frazier & Cook, 1993, used samples of less than 100), there are several concerns that should guide future research. The first is that the data were obtained from retrospective accounts of respondents and thereby subject to memory bias. Specifically, as Miell (1987) noted among intact couples, current feelings strongly influence recollections of previous relationship events. To the extent that current levels of distress color memories of the previous relationship, our findings may be an artifact of consistency in memory schema. However, our data indicated that reports of breakup distress for the period after the breakup were unaffected by length of time since the breakup, which suggests that if there is a memory bias, it is not too pervasive. Nonetheless, future research should obtain longitudinal information from both individuals who experience a breakup and should track their reactions to the course of the relationship over time. Having individuals keep detailed diaries of their breakup experiences, for example, could provide a more extensive understanding of the breakup process, and may help to curb problems of recall bias. Frequent data collection may also make it possible to obtain post-dissolution distress measures soon after the breakup, as well as providing a more accurate understanding of the causal factors associated with distress. In addition, more objective information on an individual's level of distress might be obtained by gaining the perspective of a friend or family member, or perhaps the ex-partner.

Second, many of the key constructs (e.g. satisfaction, commitment, initiation of the breakup) were assessed with one-item measures. Although such measures had some advantages (e.g. it may be easier for participants to respond, retrospectively, to global items concerning feelings prior to the breakup), in future research, more detailed measures should be used to confirm these results.

Third, the sample consists primarily of relatively young, mostly Caucasian, heterosexual and highly educated individuals. Replicating this study using respondents with varied ages, racial and ethnic backgrounds, social classes and sexual orientations would be an important contribution to the literature.

Fourth, the communicative enactment of the breakup merits more careful attention. Although the person who is left may feel some degree of rejection no matter how the breakup is accomplished, previous research suggests that some breakup strategies are more face preserving than others (Metts, 1992). Also, the extent to which partners are able to negotiate a mutually shared understanding of why the relationship has dissolved is associated with greater satisfaction with the breakup (Baxter, 1987). Thus, we would expect that the nature of the communication that occurs during the period of the breakup might well mediate some of the associations found in the current investigation.

Finally, a full understanding of the distress experienced at the time of a breakup and during the months that follow depends critically on the

inclusion of network influences. A limitation of the study reported here, and others that ignore network influences, is the implicit, although unintended, assumption that distress is an individually experienced and individually processed emotion. Compelling evidence suggests that interactions with the social network (often including the ex-partner) serve to help people reappraise the cause and intensity of their distress as well as enhance their ability to cope (Burleson & Goldsmith, 1998). As a result, variables such as time since breakup may actually have little direct influence on current distress were it not for the supportive interactions that occur during that time.

In conclusion, the loss of a close relationship can unleash a host of negative emotions. Yet we see here that the level of distress associated with relationship dissolution is not uniform; it varies from relationship to relationship and from individual to individual in predictable and systematic ways. In particular, the dissolution of relationships in which people are highly invested evokes the greatest distress, and individuals who are fearful of abandonment in the first place experience the most anguish, initially and for months following the ending of a relationship. Nevertheless, some degree of upset is likely to be associated with any breakup — a potential cost incurred by all those who engage in an intimate relationship. For most people these costs are apparently not enough to deter them from reentering, often repeatedly, into the risky but rewarding endeavor of a close, romantic liaison.

REFERENCES

- Attridge, M., Berscheid, E. & Simpson, J.A. (1995) 'Predicting Relationship Stability from Both Partners versus One', *Journal of Personality and Social Psychology* 69: 254–68.
- Bartholomew, K. & Horowitz, L.M. (1991) 'Attachment Styles among Young Adults: A Test of a Four-Category Model', *Journal of Personality and Social Psychology* 61: 226–44.
- Baxter, L.A. (1987) 'Cognition and Communication in the Relationship Process', In R. Burnett, P. McGhee, & D.D. Clarke (eds) *Accounting for Relationships*, pp. 192–212. London: Methuen.
- Berscheid, E., Snyder, M. & Omoto, A.M. (1989) 'The Relationship Closeness Inventory: Assessing the Closeness of Interpersonal Relationships', *Journal of Personality and Social Psychology* 57: 792–807.
- Birnbaum, G.E., Orr, I., Mikulincer, M. & Florian, V. (1997) 'When Marriage Breaks Up — Does Attachment Style Contribute to Coping and Mental Health?', *Journal of Social and Personal Relationships* 14: 643–54.
- Brehm, J.W. & Cohen, R.A. (1962) *Explorations in Cognitive Dissonance*. New York: Wiley.
- Burleson, B.R. & Goldsmith, D. (1998) 'How the Comforting Process Works: Alleviating Emotional Distress through Conversationally Induced Reappraisals', in P. Anderson & L. Guerrero (eds), *Communication and Emotion*, pp. 247–80. Orlando, FL: Academic Press.
- Brigotas, S. (1996, May) 'Dependence and Interaction Quality', paper presented at the Midwest Psychology Association Meetings, Chicago, IL.
- Feeney, J.A. & Noller, P. (1992) 'Attachment Style and Romantic Love: Relationship Dissolution', *Australian Journal of Psychology* 44: 69–72.
- Felmlee, D.H. (1995) 'Fatal Attractions: Affection and Disaffection in Intimate Relationships', *Journal of Social and Personal Relationships* 12: 295–311.

- Fine, M.A. & Sacher, J.A. (1997) 'Predictors of Distress Following Relationship Termination among Dating Couples', *Journal of Clinical and Social Psychology* 16: 381–8.
- Fiske, S.T. & Taylor, S.E. (1984) *Social Cognition*. Reading, MA: Addison-Wesley
- Fraley, R.C., David, K.E. & Shaver, P.R. (1997) 'Attachment Behavior and Romantic Relationship Dissolution', paper under review.
- Frazier, P.A. & Cook, S.W. (1993) 'Correlates of Distress Following Heterosexual Relationship Dissolution', *Journal of Social and Personal Relationships* 10: 55–67.
- Gray, J.D. & Silver, R.C. (1990) 'Opposite Sides of the Same Coin: Former Spouses' Divergent Perspectives in Coping with Their Divorce', *Journal of Personality and Social Psychology* 59: 1180–91.
- Hazan, C. & Shaver, P.R. (1994) 'Attachment as an Organizational Framework for Research on Close Relationships', *Psychological Inquiry* 5: 1–22.
- Hill, C.T., Rubin, Z. & Peplau, L.A. (1976) 'Breakups before Marriage: The End of 103 Affairs', *Journal of Social Issues* 32: 147–68.
- Kelley, H.H. & Thibaut, J.W. (1978) *Interpersonal Relations: A Theory of Interdependence*. New York: Wiley.
- Lazarus, R.S., Averill, J.R. & Opton, E.M. Jr (1970) 'Towards a Cognitive Theory of Emotion', in M.B. Arnold (ed.) *Feelings and Emotions: The Loyola Symposium*, pp. 207–32. New York: Academic Press.
- McIntosh, W.D. & Martin, L.L. (1992) 'The Cybernetics of Happiness: The Relation of Goal Attainment, Rumination, and Affect', in M.S. Clark (ed.) *Emotion and Social Behavior*, pp. 222–46. Newbury Park, CA: Sage.
- Metts, S. (1992) 'The Language of Disengagement: A Face-management Perspective', in T.L. Orbuch (ed.) *Close Relationship Loss*, pp. 111–27. New York: Springer-Verlag.
- Metts, S., Cupach, W.R. & Bejlovec, R.A. (1989) '"I Love You Too Much To Ever Start Liking You": Redefining Romantic Relationships', *Journal of Social and Personal Relationships* 6: 259–74.
- Miell, D. (1987) 'Remembering Relationship Development: Constructing a Context for Interaction', in R. Burnett, P. McGhee & D. Clarke (eds) *Accounting for Relationships: Explanations, Representation and Knowledge*, pp. 60–73. London: Methuen.
- Mikulincer, M., Florian, V. & Weller, A. (1993) 'Attachment Styles, Coping Strategies, and Posttraumatic Psychological Distress: The Impact of the Gulf War in Israel', *Journal of Personality and Social Psychology* 54: 817–26.
- Rusbult, C.E. (1983) 'A Longitudinal Test of the Investment Model: The Development (and Deterioration) of Satisfaction and Commitment in Heterosexual Involvements', *Journal of Personality and Social Psychology* 45: 101–17.
- Simpson, J.A. (1987) 'The Dissolution of Romantic Relationships: Factors Involved in Relationship Stability and Emotional Distress', *Journal of Personality and Social Psychology* 53: 683–92.
- Simpson, J.A. (1990) 'Influence of Attachment Styles on Romantic Relationships', *Journal of Personality and Social Psychology* 59: 971–80.
- Sprecher, S. (1994) 'Two Sides to the Breakup of Dating Relationships', *Personal Relationships* 1: 199–222.